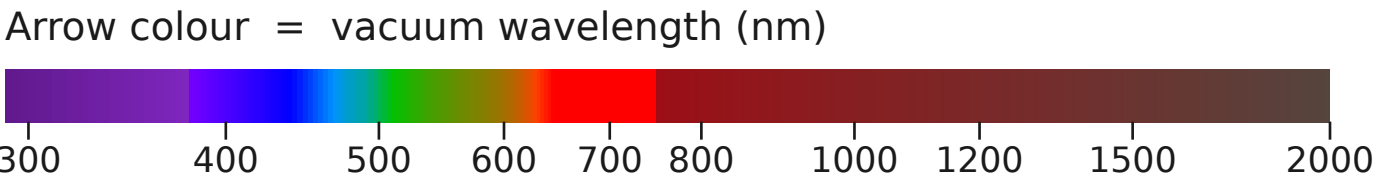


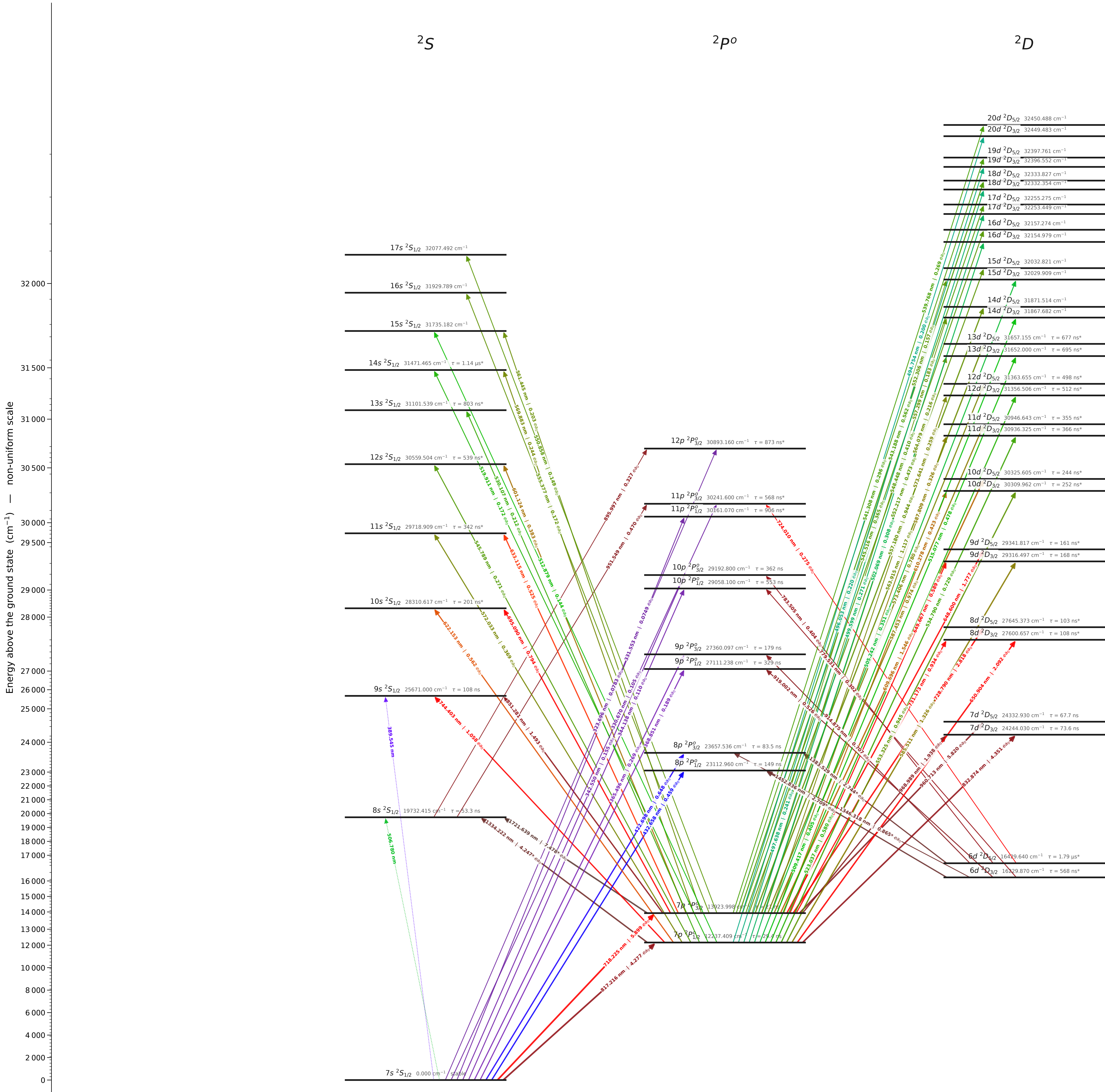
# Francium

Arrow label: vacuum wavelength | reduced dipole matrix element  $\langle J||e\mathbf{r}||J'\rangle$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Fr (Fr I). Strongest classified transitions below 2 μm (NIST): 52 levels, 77 lines.

Fr<sup>+</sup> ionisation limit 32848.87 cm<sup>−1</sup> = 4.07274 eV (304.424 nm)



## Key transitions

| Transition                                                                                      | λ vac (nm) | λ air (nm) | ν (THz)      | $\langle J  e\mathbf{r}  J'\rangle$ ( $e a_0$ ) | A (s <sup>−1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|-------------------------------------------------------------------------------------------------|------------|------------|--------------|-------------------------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>9s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 389.5446   | 389.4343   | 769.5972     | –                                               | –                    | 1.48           | –          | two-photon 75-9S spectroscopy of trapped 210Fr               |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>8p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 422.6983   | 422.5793   | 709.23507(3) | 0.648                                           | 2.820e+06            | 1.906          | 45.3       | blue line                                                    |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>8p</i> <sup>2</sup> <i>P</i> <sub>1/2</sub> | 432.6577   | 432.5361   | 692.9091     | 0.459                                           | 2.640e+06            | 1.066          | 17.4       | blue line                                                    |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>8s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 506.7803   | 506.6391   | 591.5629     | –                                               | –                    | 2.986          | –          | two-photon / parity-violation transition 7S-8S               |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.2251   | 718.0272   | 417.40739(6) | 5.899                                           | 4.757e+07            | 7.572          | 100        | D2 line                                                      |
| <i>7p</i> <sup>2</sup> <i>P</i> <sub>1/2</sub> – <i>9s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 744.4026   | 744.1976   | 402.7289     | 1.058                                           | 2.750e+06            | 1.48           | 27.2       | 744.4 nm: decay channel of 9S                                |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>1/2</sub> | 817.2165   | 816.9918   | 366.84584(6) | 4.277                                           | 3.396e+07            | 5.404          | 100        | D1 line                                                      |
| <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> – <i>7d</i> <sup>2</sup> <i>D</i> <sub>3/2</sub> | 832.8738   | 832.6449   | 359.9494     | 4.351                                           | 1.660e+07            | 2.162          | 78.8       | 832.9 nm: 7P1/2 - 7D3/2                                      |
| <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> – <i>9s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 851.2810   | 851.0472   | 352.1663     | 1.493                                           | 3.660e+06            | 1.48           | 34.1       | 851.3 nm: 7P3/2 - 9S step                                    |
| <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> – <i>7d</i> <sup>2</sup> <i>D</i> <sub>5/2</sub> | 960.7134   | 960.4499   | 312.0519     | 5.820                                           | 1.290e+07            | 2.351          | 99.5       | 960.7 nm: 7P3/2 - 7D5/2                                      |
| <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> – <i>7d</i> <sup>2</sup> <i>S</i> <sub>3/2</sub> | 968.9892   | 968.7236   | 309.3868     | 1.938                                           | 2.090e+06            | 2.162          | 19.3       | 969.0 nm: 7P3/2 - 7D3/2                                      |
| <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> – <i>8s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 1334.2218  | 1333.8570  | 224.6946     | 4.247*                                          | 7.693e+06            | 2.986          | 40.9       | 1.33 um: second step of the two-photon 7S-8S excitation u... |
| <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> – <i>8s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 1721.6395  | 1721.1694  | 174.1320     | 7.478*                                          | 1.110e+07            | 2.986          | 59.1       | 1.72 um: decay channel of 8S                                 |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## <sup>212</sup>Fr hyperfine structure (I = 5)

| Level                                           | A (MHz)     | B (MHz)    | Shift of each F level from the centre of gravity (MHz)                                  |
|-------------------------------------------------|-------------|------------|-----------------------------------------------------------------------------------------|
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub>  | 9064.20(20) | –          | F=9/2: -27192.60 F=11/2: +22660.50                                                      |
| <i>7p</i> <sup>2</sup> <i>P</i> <sub>1/2</sub>  | 1192.00(20) | –          | F=9/2: -3576.00 F=11/2: +2980.00                                                        |
| <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub>  | 97.21(10)   | -26.00(20) | F=7/2: -886.16 F=9/2: -431.81 F=11/2: +107.61 F=13/2: +722.58                           |
| <i>8p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub>  | 373.00(10)  | –          | F=9/2: -1119.00 F=11/2: +932.50                                                         |
| <i>8p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub>  | 32.80(10)   | -7.7(9)    | F=7/2: -298.54 F=9/2: -145.93 F=11/2: +35.88 F=13/2: +244.07                            |
| <i>8d</i> <sup>2</sup> <i>D</i> <sub>3/2</sub>  | 13.0(6)     | –          | F=7/2: -117.00 F=9/2: -58.50 F=11/2: +13.00 F=13/2: +97.50                              |
| <i>8d</i> <sup>2</sup> <i>D</i> <sub>5/2</sub>  | -7.2(6)     | –          | F=5/2: +108.00 F=7/2: +82.80 F=9/2: +50.40 F=11/2: +10.80 F=13/2: -36.00 F=15/2: -90.00 |
| <i>10s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 401(5)      | –          | F=9/2: -1203.00 F=11/2: +1002.50                                                        |
| <i>9d</i> <sup>2</sup> <i>D</i> <sub>3/2</sub>  | 7.1(7)      | –          | F=7/2: -63.90 F=9/2: -31.95 F=11/2: +7.10 F=13/2: +53.25                                |
| <i>9d</i> <sup>2</sup> <i>D</i> <sub>5/2</sub>  | -3.6(4)     | –          | F=7/2: +54.00 F=9/2: +41.40 F=11/2: +25.20 F=13/2: +5.40 F=15/2: -18.00 F=17/2: -45.00  |
| <i>11s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 225(3)      | –          | F=9/2: -675.00 F=11/2: +562.50                                                          |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                                                                                      | λ vac (nm) | Isotopes | Shift (MHz)  |
|-------------------------------------------------------------------------------------------------|------------|----------|--------------|
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>8p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 422.698    | 213-212  | -1615.0(3)   |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>8p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 422.698    | 220-212  | -20564.0(7)  |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>8p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 422.698    | 221-212  | -23298.0(8)  |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>8s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 506.780    | 213-208  | -5123(6)     |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>8s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 506.780    | 213-209  | -3678(6)     |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>8s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 506.780    | 213-210  | -3274(6)     |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>8s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> | 506.780    | 213-211  | -1958(6)     |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 207-212  | 5239(4)      |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 208-212  | 5003(3)      |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 209-212  | 3133.0(20)   |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 210-212  | 2603.0(10)   |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 211-212  | 901(3)       |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 222-212  | -26262(3)    |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 223-212  | -27922.0(20) |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 224-212  | -30891.0(10) |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 225-212  | -32297(6)    |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 226-212  | -34401.0(10) |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 227-212  | -38352.0(20) |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 228-212  | -40077(5)    |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 221-212  | -23570.0(20) |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 213-212  | -1641(3)     |
| <i>7s</i> <sup>2</sup> <i>S</i> <sub>1/2</sub> – <i>7p</i> <sup>2</sup> <i>P</i> <sub>3/2</sub> | 718.225    | 220-212  | -20806.8(5)  |

Shift = ν(first isotope) – ν(second isotope).